UNLV | Howard R. Hughes College of Engineering

Open-Plan Three-Way Speaker Design

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Design Goals

My idea is inspired by the uprising trend in modern living spaces called an open-plan concept, in which I noticed my home having its own version of this with a shared living room, kitchen, and dining area. Therefore, I formed a concept to design a speaker that would work in an open space, such as my home's downstairs living area.

The goal of this speaker is for its sound to be evenly distributed and heard well throughout the entire space, whether you are cooking in the kitchen, eating at the dining table, or sitting in the living area. I will have to figure out which drivers would be ideal for a small, yet open space, what box shape works best with those drivers, and where the speaker will be placed.

After researching speakers in different spaces, such as in movie theatres and in homes, I concluded that a three-way speaker with good lower end audio is a top priority within my design. As this speaker will be connected to the TV in my living room, I want something that can handle the wide range of audio that films provide. An additional use would be for playing pop music, and it would be beneficial to have those bass notes. Another benefit to designing a speaker for my home is that the smaller space offers more precise audio with less distortion compared to a movie theatre where audio distortion is more apparent.

Driver Choice

Woofer

Considering my main design goal was to create a speaker with deep bass and clear audio, I began my driver selection process with the woofer. Using PartsExpress.com, I narrowed down my search to look for 10" or 12" woofers, and I ended up finding a wide variety of drivers for different uses, most of which being from the brands Dayton Audio and GRS. I also included a filter for an impedance of 8Ω as opposed to 4Ω, considering that it is safer for home audio speakers because an amplifier would not be needed to avoid overheating.

My first option is the Dayton Audio DC250-8 10" Classic Woofer, which offers an 8Ω impedance, a frequency response of 25 to 2,500 Hz, a Qts of 0.38, a power handling of 70 to 140 Watts, and a sensitivity of 88.4dB at 2.83V/1m. The description also mentions that this woofer is ideal for two-way or three-way speaker designs and can provide the versatility of being in either a sealed or ported enclosure due to its Qts. Additionally, this frequency range tells me that this woofer will provide the low-end sounds I am looking for, putting this driver as one of my top options.

My second option is the GRS 10" Woofer Surface Mount Poly Cone 8 Ohm 10SMP-8. With an impedance of 8Ω, a frequency response of 24 to 2,000 Hz, a Qts of 0.58, a power handling of 70 to 140 Watts, and a sensitivity of 87.5db at 2.83V/1m, this woofer is a competitive contender to my first option. It has a lower frequency response, keeps the same impedance, size, and power handling, and has a lower sensitivity at 87.5dB compared to 88.4dB at 2.83V/1m. Its Qts is higher at 0.58 compared to 0.33, meaning it would be better optimized for sealed enclosures. Also, the lower frequency response would be better suited for a three-way speaker design.

After careful consideration of these woofer options for my speaker design, I landed on the Dayton Audio DC250-8 10" Classic Woofer as it provides the features that I am looking for, including the low frequency response and higher sensitivity. While the GRS woofer was a close choice, the Dayton Audio woofer provides more versatility with the enclosure and has a slightly higher sensitivity, making it require less power to be slightly louder. This would be better for the space I am making this speaker for as it would be able to reach further areas like the kitchen and dining area. Additionally, when looking at the frequency response charts of both, the Dayton

woofer has a smoother roll-off due to its paper cone, making it easier to integrate the mid-range driver.

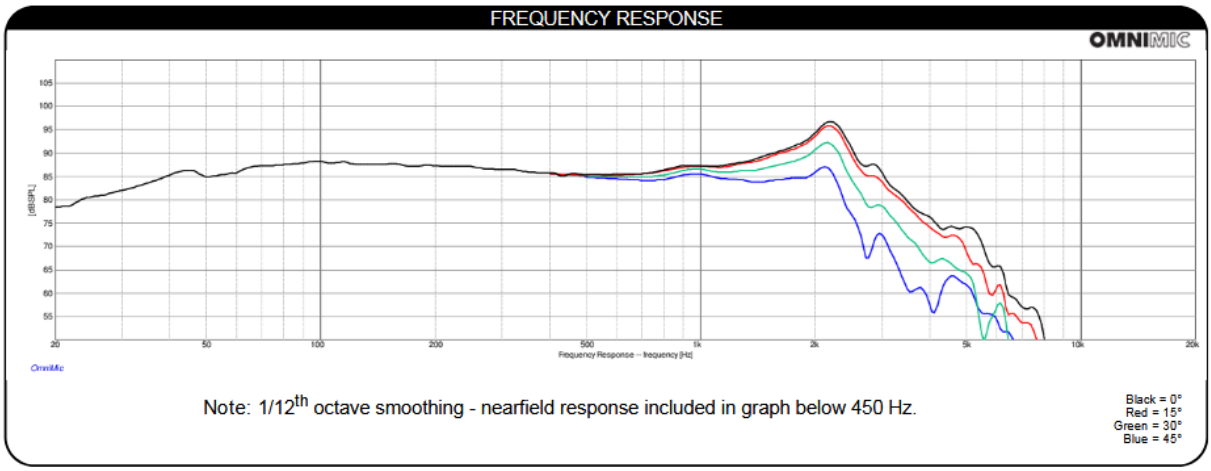


Table 1: Woofer frequency response chart

Midrange

For my midrange driver, I used PartsExpress.com again and narrowed down my search to 5” or 5-1/4” drivers, as those specific sizes had more options to compare with. While searching, I found out that Dayton Audio made multiple drivers in the same “Classic” series, one being the Dayton Audio DC130B-8 5-1/4" Classic Woofer Speaker. It has an 8Ω impedance, a frequency response of 50 to 5,000 Hz, a Qts of 0.38, power handling of 40 to 80 Watts, and a sensitivity of 89.1dB at 2.83V/1m. This midrange has the same paper cone that the woofer has, making them easy to blend together. Its sensitivity is very close to the 88.4dB sensitivity of my woofer, and the frequency response is perfect for vocal sounds in films and music.

My other option for a midrange driver is the Dayton Audio RS125P-8 5" Reference Paper Woofer 8 Ohm. It has an 8Ω impedance, a frequency response of 70 to 10,000Hz, a Qts of 0.44, power handling of 30 to 45 Watts, and a sensitivity of 87.1db at 2.83V/1m. This driver has a

much higher frequency response range, which would make it easier for my crossover between the midrange and tweeter.

Overall, I decided to go with the first option from the same Classic series, the Dayton Audio DC130B-8 5-1/4" Classic Woofer Speaker. While both midrange drivers had pros to them, I decided on the same series midrange as the similar cones provided smooth roll-offs again. Also, this Classic driver has a higher sensitivity of 89.1dB at 2.83V/1m, compared to the second option at 87.1dB, meaning the louder sound would help fill the larger room.

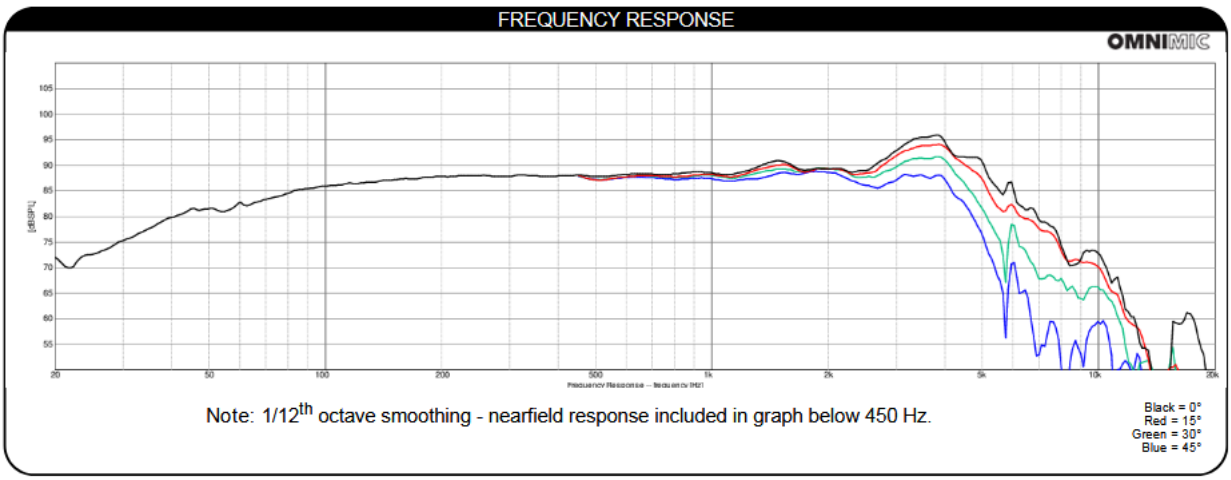


Table 2: Midrange frequency response chart

Tweeter

As both my woofer and midrange drivers were from Dayton Audio, I decided that the tweeter should also be from the same brand to help make the crossover easier. I narrowed down my search to Dayton Audio tweeters that had an impedance of 8Ω to keep my drivers consistent.

My first option is the Dayton Audio DC25T-8 1" Titanium Dome Tweeter. It has the 8Ω impedance needed, a frequency response of 3,000 to 20,000 Hz, a Qts of 0.47, power handling of 50 Watts, and a sensitivity of 93dB at 2.83V/1m. This frequency response range is quite high, starting at 3,000 Hz, which may make it more difficult to crossover between the midrange driver.

Additionally, the sensitivity of 93dB is too high compared to my other driver choices, requiring a resistor to prevent the high frequencies from overpowering my woofer's bass.

My second option is the Dayton Audio DC28F-8 1-1/8" Silk Dome Tweeter. It has an 8Ω impedance, a frequency response of 1,300 to 20,000 Hz, a Qts of 0.5, power handling of 50 Watts, and a sensitivity of 89dB at 2.83V/1m. Right away, I noticed that this tweeter has a wider frequency response range that starts at 1,300 Hz as opposed to 3,000 Hz, allowing me to have a wider crossover window to play around with while protecting my midrange driver from working too hard. Additionally, this tweeter's 89dB sensitivity is very close to the 88.4dB sensitivity of the woofer and 89.1dB sensitivity of the midrange driver, making it easier to match sensitivities in my crossover.

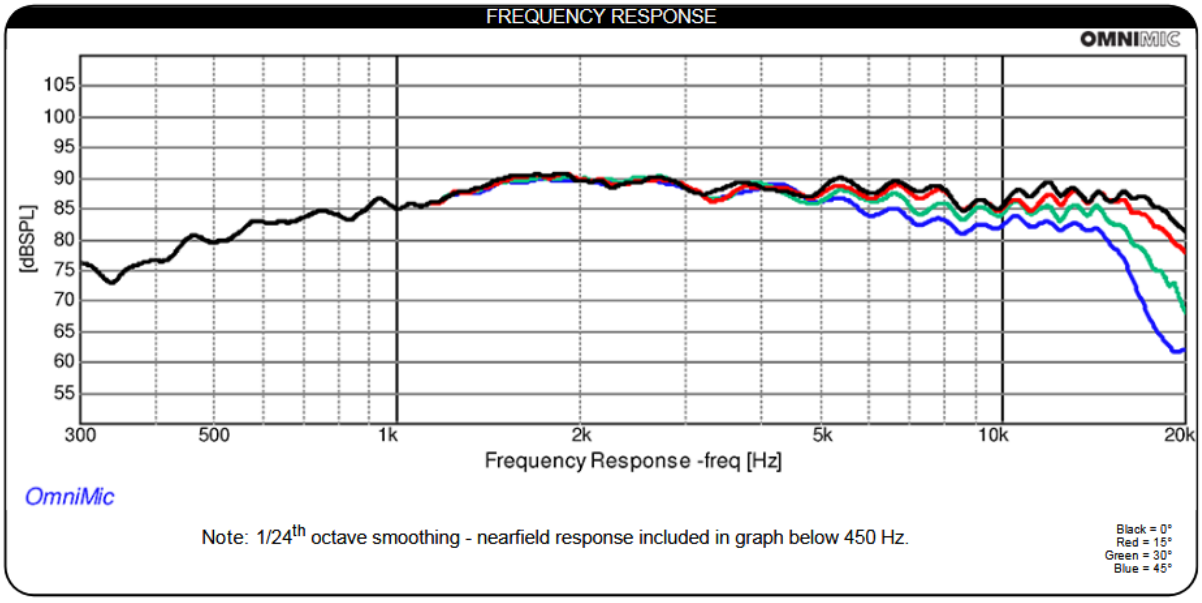


Table 3: Tweeter frequency response chart

Driver Conclusion

Overall, I am very satisfied with my choices as I gave myself enough range between each driver to play around with where I would want the sounds to cross over, and it turned into a

smooth graph! My choices will absolutely serve my original design goal of having a low-end focused speaker that can be heard from across an open room, considering that the average sensitivity of all three of my drivers is 88.83dB at 2.83V/1m. Regarding potential issues with my box, I may have difficulty figuring out whether or not I want a sealed or ported enclosure, and, if ported, what kind of port do I want to use and how long that would need to be.

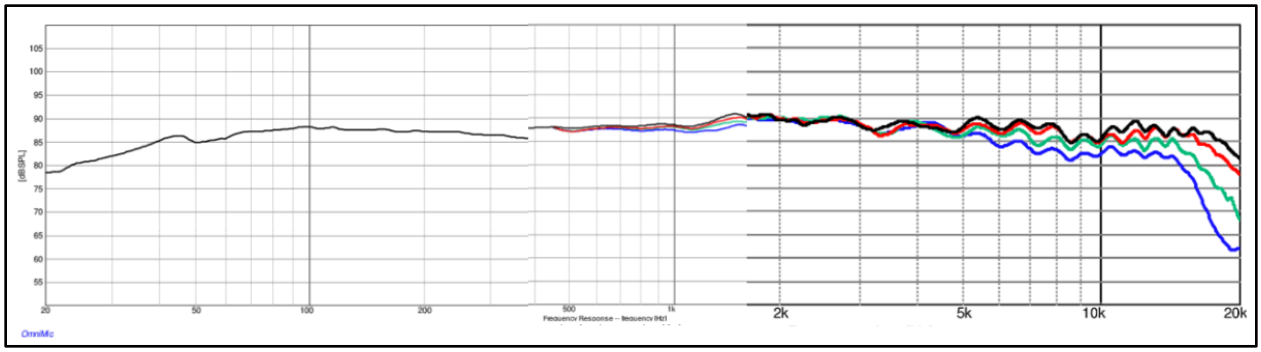


Table 4: Overall frequency response chart

Box Design

Ported vs Sealed.

When considering my box design, I must first decide between a ported or sealed box. Each type of box design has its own pros and cons. A ported box offers louder bass without the use of additional power. However, a larger box size is needed, and the sound is less accurate as audio waves bounce between the walls of the port. A sealed box offers tighter and more accurate bass in a smaller box. However, more power would be needed for a louder sound with a sealed box. To figure out which type of box is needed for my design, I use Equation 1 to calculate my Efficiency Bandwidth Product (EBP) and plug in my woofer's resonant frequency (Fs) and electromagnetic Q (Qes) values. If my EBP is around 50 or less, then the woofer is suited for a

sealed box. If the EBP is greater than 100, then the woofer is suited for a ported box. If the EBP is between 50 to 100, then either box works.

$$EBP = \frac{Fs}{Qes} \quad (1)$$

Plugging in my woofer's Thiel-Small parameters, 24.9 Hz divided by 0.44 equals an EBP of 56.59, meaning either a ported or sealed enclosure would work. Additionally, the description on the woofer itself states that the Qts of 0.38 allows it to work well in both sealed and ported speaker enclosures, confirming my calculations. As a reminder, my main design goal is to create a speaker that works well in an open space, while also fitting in a living room next to a TV. Since I would use my speaker for pop music and movies, I would prefer to have a sealed box for more accurate bass sounds and smaller size.

Volume of the Box

The volume of my sealed box is calculated using Equation 2 and plugging in my woofer's Thiel-Small parameters once more. The woofer's Compliance Equivalent Volume (Vas) is 3.67 ft^3 and its Total Q (Qts) is 0.38. As for my desired alignment, or the Qtc value, I am considering both a critically damped and a Butterworth alignment. The main benefit of a critically damped alignment is its excellent transient response, meaning the speaker would deliver a clear, detailed sound without adding distortion. However, this kind of alignment would result in a box that is 2 to 3 times larger than a Butterworth box. A Butterworth alignment offers a balanced sound with a smaller enclosure, but it is underdamped in comparison, resulting in a slight ringing sound and less detailed bass sounds.

$$Vb = \frac{Vas}{\left(\frac{Qtc}{Qts}\right)(2) - 1} \quad (2)$$

To consider both options, I have calculated the box volume with both a critically damped Qtc of 0.5 and a Butterworth Qtc of 0.707 using Equation 2. The box volume with Qtc 0.5 is 2.249 ft^3 or 3886 in^3, and the box volume with Qtc 0.707 is 1.349 ft^3 or 2331 in^3.

With a critically damped alignment, my box size would be 24" x 16" x 10", which comes out to 3840 in^3. With a Butterworth alignment, my box size would be 20" x 13" x 9", or 2340 in^3. Considering this calculated box size is specifically for just the woofer's desired response, my speaker would be taller than these dimensions to account for my midrange driver and tweeter. While I would prefer to use a critically damped alignment due to its amazing transient response, the box volume is too big for the space of my living room, so my final choice for alignment is Butterworth.

To determine the resonant frequency of the box, I use Equation 3 and plug in my calculated box volume and my woofer's Thiel-Small parameters once again. With a resonant frequency (Fs) of 24.9 Hz, a Compliance Equivalent Volume (Vas) of 3.67 ft^3, and a box volume (Vb) of 1.349 ft^3, the calculated resonant frequency of the box (Fb) is 48.03 Hz.

$$Fb = Fs\sqrt{\left(\frac{Vas}{Vb} + 1\right)} \qquad 3$$

Sealed

As I have chosen a sealed design as opposed to a ported design, I would not need to use Equation 4 to calculate my port length. Since I chose a sealed enclosure, the sound of my speaker will be more articulate, faster, and accurate to the original sound it is playing compared to if I had chosen a ported enclosure. While I would need more power for my speaker to be louder due to the smaller box and lack of a port, I believe my speaker is perfectly loud enough with the drivers alone considering it will be used in my open living room space.

$$Lv = \frac{23562.5 * Dv^2 * Ne}{F6^2 * V6} - (K * Dv) \quad (4)$$

Frequency response chart

Plugging in the woofer's Thiel-Small parameters, calculated box volume (Vb), and the Butterworth (Qtc) value resulted in the sealed enclosure frequency response graph shown in Table 5.

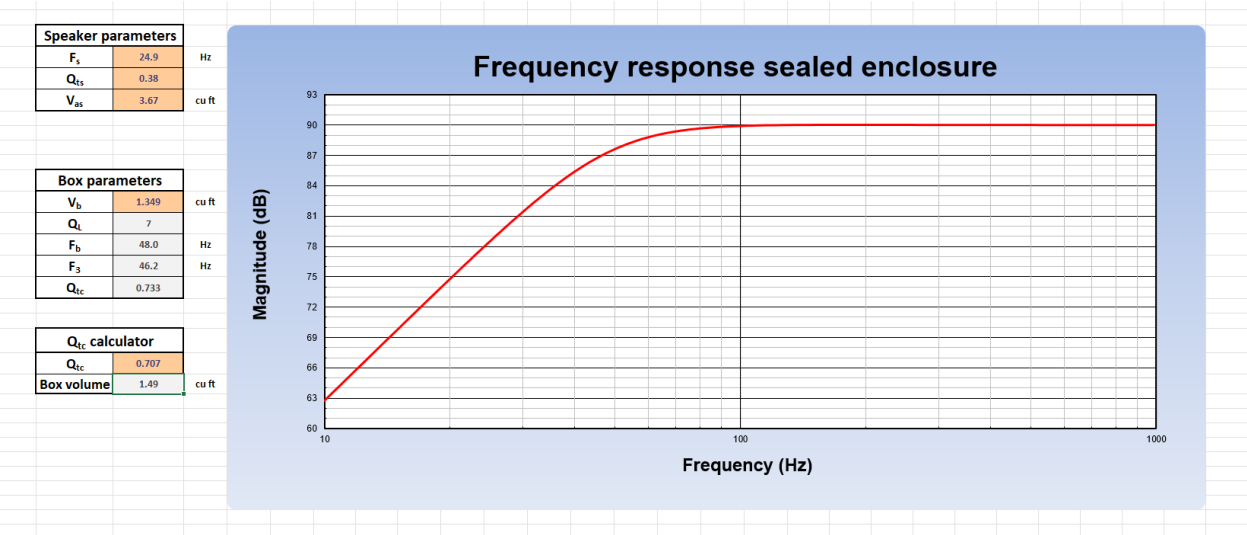


Table 5: Frequency response sealed enclosure in imperial units

From Table 5, we see that at approximately 100 Hz, the magnitude peaks at 90 dB. The line has a typical Butterworth curve and flatness.

Drawings

For the material of my speaker enclosure, I am using ¾" (19mm) high-density fiberboard (HDF) to reduce resonance and keep structural rigidity. This decision came following a conversation with my coworker at Allegiant Stadium, Andrew Paonessa, who is an experienced audio technician and the current manager of production technology. On Friday, April 10, 2026, we discussed which materials I could use and what kind of box I wanted my speaker to look like. My design is inspired by an old speaker he thrifted that sits in his office called the Pioneer CS-

M555, shown in Figure 1. It is a 3-way speaker system with a frequency range of 32 to 20,000 Hz, an impedance of 8Ω, and a 10” subwoofer. This speaker sounded amazing for what I wanted and happened to have the same impedance and subwoofer size as my speaker. It also has a similar frequency range to my speaker, which is 25 to 20,000 Hz. The material of the CS-M555 speaker is HDF, so I would like to use that for my speaker design as well. The main difference between this speaker and mine is that it is ported.



Figure 1: Pioneer CS-M555 Speaker

I used a ¾” thickness for the HDF in my model, with 21” x 11” x 10-1/8”, or 2339 in^3, inner dimensions for the woofer box. My 10” woofer has a baffle cutout diameter of 9.21”, so I modeled that cutout as well. After researching midrange driver and tweeter placements, I found out that since my tweeter has a silk dome, it does not need its own box because it is self-sealed.

For my midrange driver, which is 5 ¼” with a baffle cutout diameter of 4.72”, I will be using box dimensions of 10.45” x 11” x 10-1/8”, or 1164 in^3. Additionally, my tweeter has a baffle cutout diameter of 2.91”. I modeled my 3-way speaker box design using Fusion 360, as shown in Figures 2, 3, 4, and 5.



Figure 2: Isometric view of 3-way speaker box design

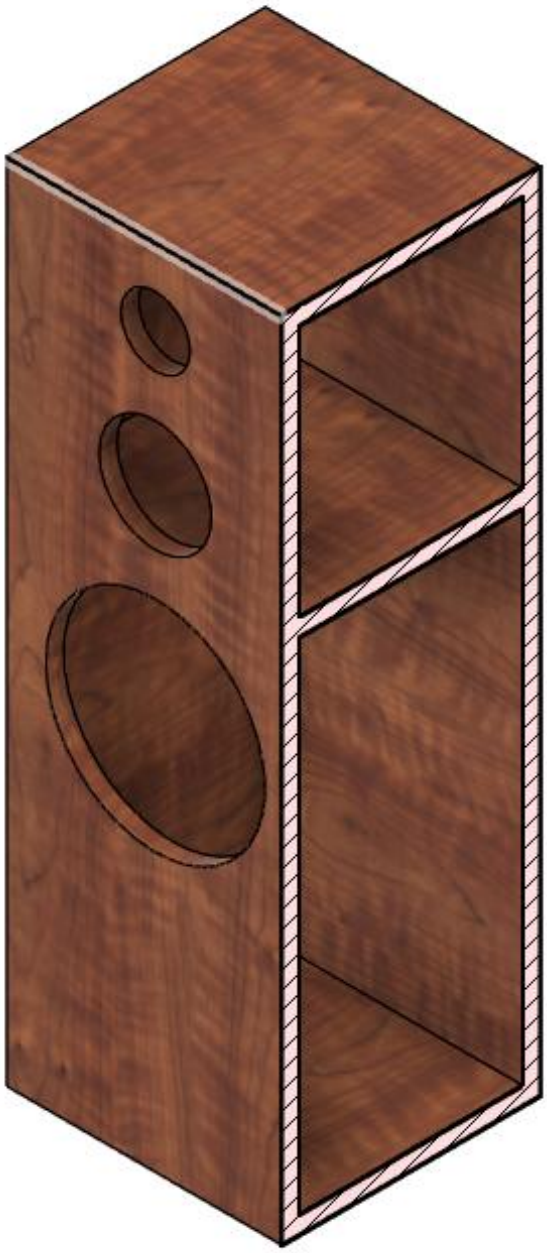


Figure 3: Isometric view with section analysis

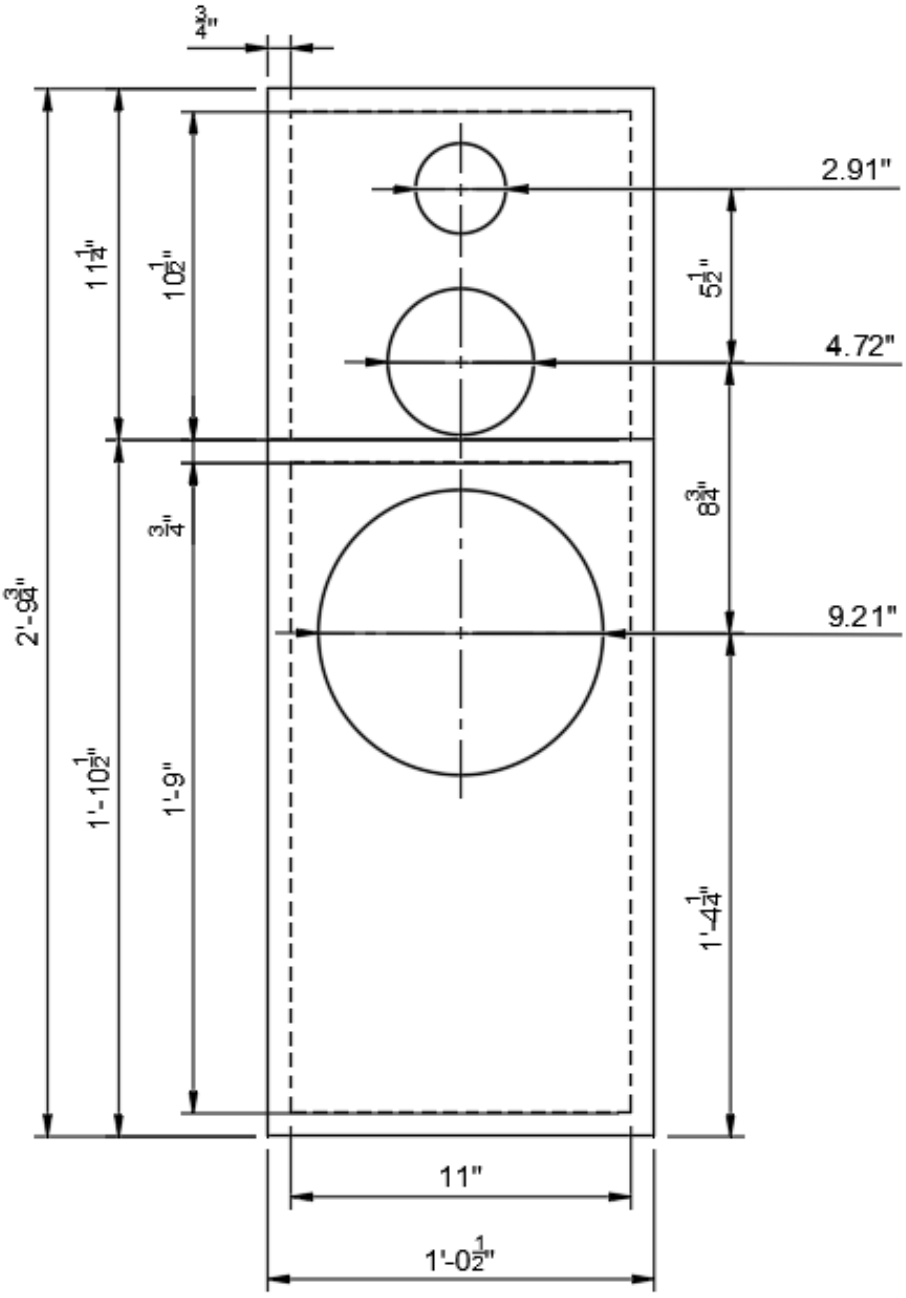


Figure 4: Front view with dimensions

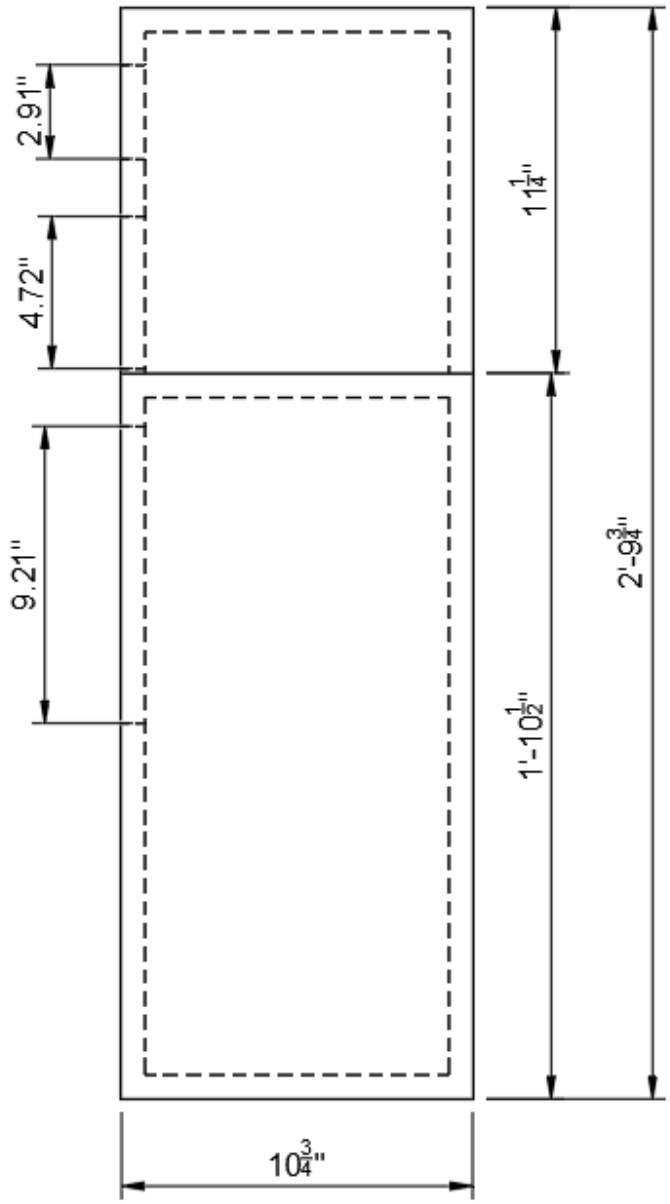


Figure 5: Side view with dimensions

The overall specs of my speaker are 2'-9¾" x 1'-½" x 10¾". The woofer is placed on the bottom with the midrange driver directly above it in a separate box. There is a ¾" thick shelf between the midrange driver box and the woofer box to keep the bass sounds from leaking, sounding muddy, and damaging the midrange driver. The tweeter is placed above the midrange

driver so the sounds can mesh well. My design takes direct inspiration from the Pioneer speaker in Figure 1, as well as the classic floor-standing tower speaker design.

Crossovers

Inductors and Conductors

While I have my three drivers selected, there are additional steps needed to optimize each driver to a certain frequency range. The last step I need to complete is creating a crossover to prevent my speaker's audio from sounding muddy. We have the inductor, which allows low frequencies to pass while blocking high frequencies. I will be using an inductor for my woofer as a low-pass filter. A conductor does the opposite; it lets high frequencies pass while blocking low frequencies, acting as a high-pass filter for my tweeter. For my midrange driver, I will be using both an inductor and a conductor to act as a band-pass filter. For my speaker, I am aiming for the crossover between the woofer and midrange driver to be at 500 Hz and the midrange driver and tweeter crossover to be at 2000 Hz, which are typical crossover values. It works with my driver selections because my woofer, midrange driver, and tweeter's ranges naturally give me a smooth blend to overlap my crossovers.

To perfect my speaker's crossover, I calculated the inductance (L) and capacitance (C) of the crossover parts using equations 5 and 6. Calculating the inductance and capacitance of the crossover parts, I plug in the impedance of 8Ω (R), the desired crossover frequency of 500 Hz for (f), and the quality factor desired (q). For my crossover, I want a Linkwitz-Riley filter (q = 0.5) so my drivers can blend sounds together smoothly at the crossover without resulting in any extra volume. With these values, the inductance of my woofer low-pass filter comes out to L = 5.09

mH, and my midrange band-pass filter comes out to C = 19.9 µF. Now, plugging in the same values and 2000 Hz for (f) instead, we get L = 1.27 mH for my midrange band-pass filter, and C = 4.97 µF for my tweeter high-pass filter. With the three of my drivers, I now get a 3-way crossover.

$$L = \frac{R}{(2\pi)(f)(q)} \tag{5}$$

$$C = \frac{q}{(2\pi)(f)(R)} \tag{6}$$

Using VituixCAD, I created a basic crossover circuit using the values calculated and the specifications from each page of my three drivers, as shown in Figure 6. I modeled it after a 2nd-order Linkwitz-Riley style crossover.

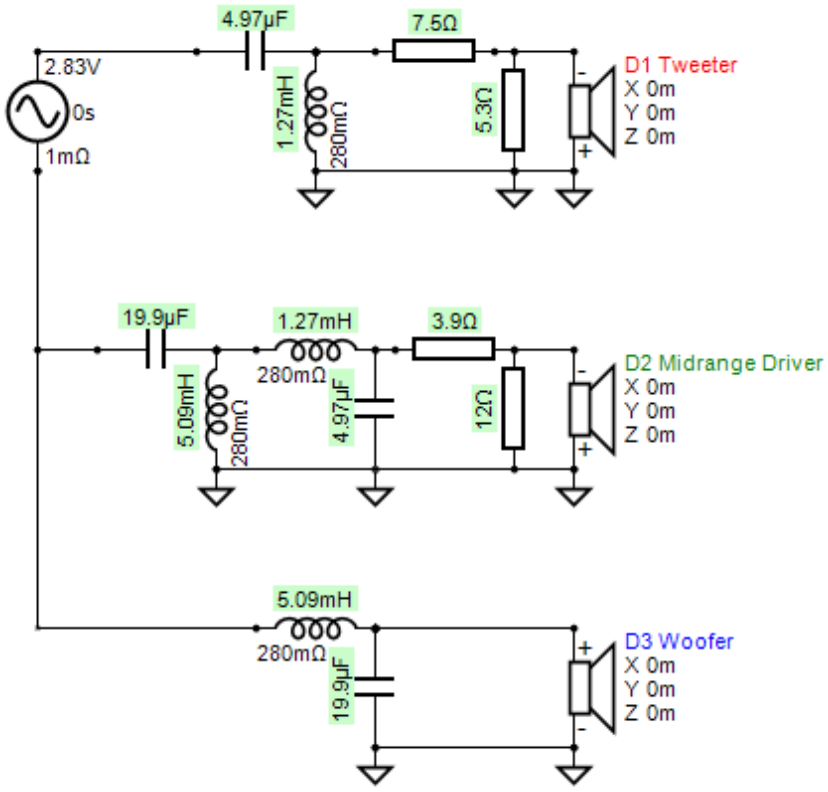


Figure 6: Circuit diagram

I inverted my tweeter and midrange polarity to prevent a dip at the 500 Hz crossover frequency and fix my phase alignment. I found that even though my three drivers had similar sensitivities of around 89 dB, the VituixCAD simulation showed that the tweeter and midrange drivers were louder due to the crossover overlap and ear sensitivity to high pitch noises. Therefore, I added two L-pad resistors to my circuit in front of my tweeter and midrange driver to help flatten the crossover. Using Equations 7 and 8, I calculated the L-pad resistor values for the resistor parallel to the speaker (Rp) and the resistor in series with the speaker (Rs). The value (k) was calculated by finding the difference in decibels between the drivers.

$$Rp = Re\frac{k}{1-k} \quad 7$$

$$Rs = Re(1-k) \quad 8$$

SPL Graphs

The tweeter is the red line, the midrange driver is the green line, and the woofer is the blue line. There is a slight dip in the 1000 Hz range where the woofer and midrange driver are crossing over, but I adjusted the L-pad values to accommodate for that as best as it could. The line is relatively flat with just two peaks and one dip, but I consider this crossover a success.

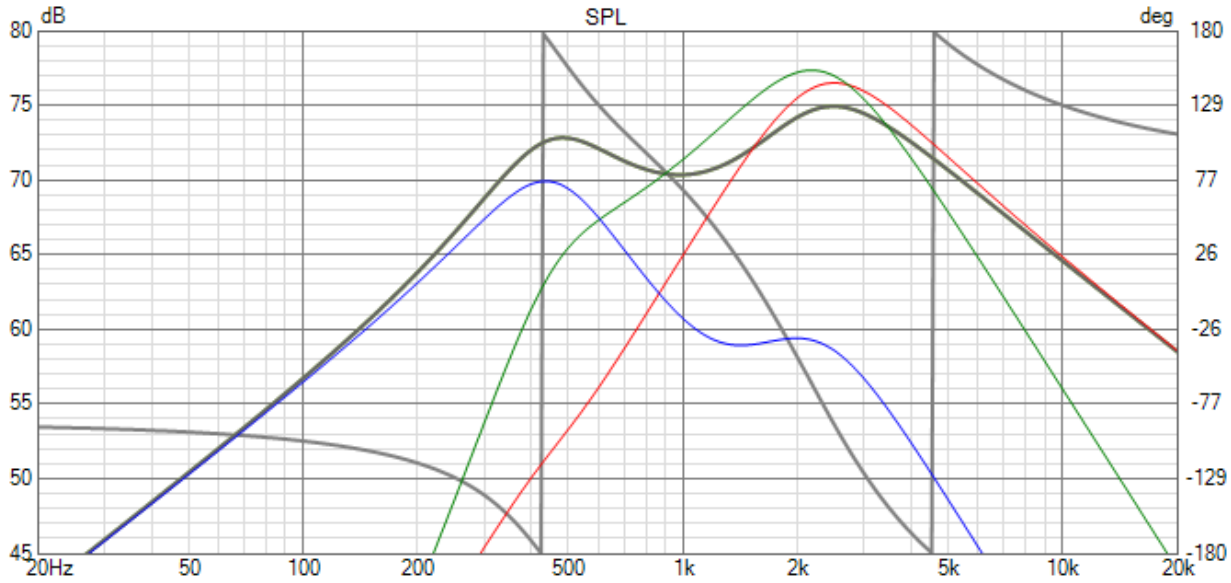


Table 6: Crossover SPL chart

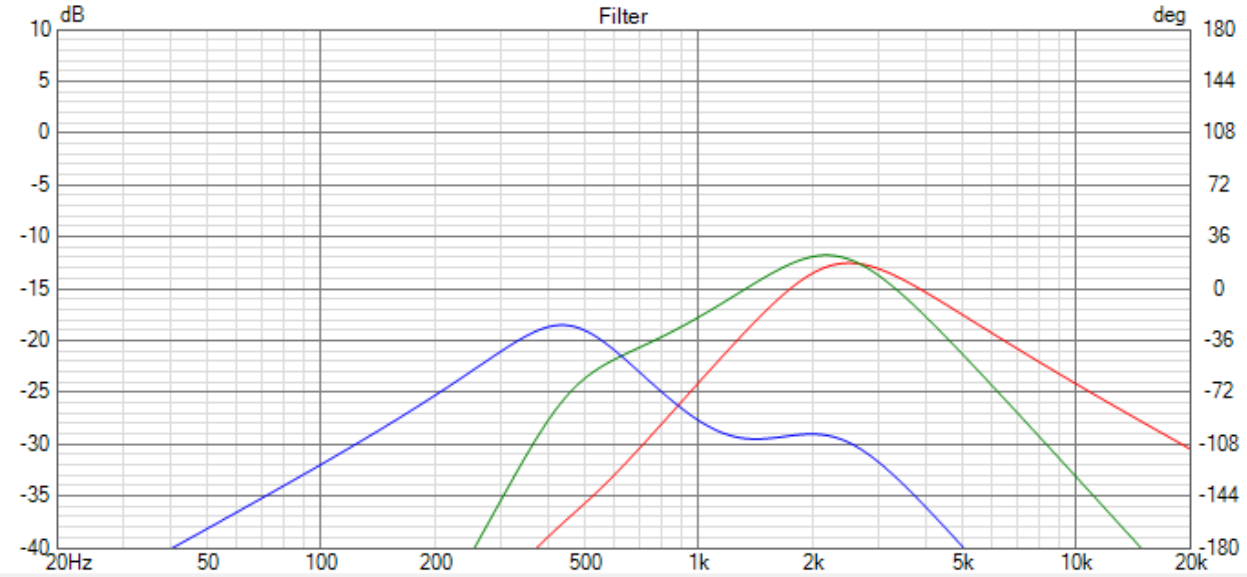


Table 7: Crossover filter chart

Speaker Conclusion

My final three-way speaker design successfully achieved all the goals I set for it. I wanted the speaker to sound evenly distributed and heard well throughout the entire space of my open living area. I researched and chose drivers that were ideal for a small, yet open space, and I 3D modeled and designed a speaker box shape that works best with my three drivers. The box is simple enough to build, while also helping amplify the sound even louder. It looks aesthetically pleasing, and it is compact at only about 1' wide and less than 3' tall, meaning it could easily fit in my living room.

The cost to build my speaker includes the three drivers, the cabinet material, and the crossover components which include the capacitors, inductors, and resistors. The three drivers cost $100.94 altogether. The cabinet material costs $128.50 for a ¾" x 4' x 8' sheet of high-density fiberboard (HDF) from Baker Lumber Co. However, upon further research, I have decided to go with ¾" x 4' x 8' sheet of medium-density fiberboard (MDF) from Home Depot instead. This reduces the cabinet cost to $49.98.

The capacitors, inductors, and resistors purchased would all be from Dayton Audio to work best with my Dayton Audio drivers. The following cost estimates for the capacitors are as follows. For my tweeter and midrange driver, two polypropylene capacitors of 5.1 µF would cost

$3.58. For my midrange driver and woofer, two polypropylene capacitors of 20 μF would cost $10.18. The following cost estimates for the inductors are as follows. For my tweeter and midrange driver, two 1.2 mH air core inductors would cost $13.58. For my midrange driver, one 5.0 mH air core inductor would cost $20.29. For my woofer, one 5.0 mH I-core inductor would cost $12.19. The following cost estimates for the resistors are as follows. For my tweeter L-pad resistors, one DNR-7.5 Ω and one DNR-5.1 Ω resistor would cost $2.50 together. For my midrange woofer L-pad resistors, one DNR-4.0 Ω resistor and one DNR-12.5 Ω resistor would cost $2.50 together. The total cost for all capacitors, inductors, and resistors in my crossover circuit is $64.82.

The overall cost of my speaker comes out to $215.74. This is a reasonable price for a speaker, and a good amount of the cost was cut since I decided to use MDF instead of HDF. If it were my original box design with the HDF, the speaker would cost $294.26 to build, which is $78.52 more. While either type of wood would work for my design, MDF, in this case, provides more benefits as it is cost effective, readily available, and easier to work with. Additionally, despite MDF being more sensitive to moisture, the environment of my home and my city is dry throughout most of the year, so that would not be an issue regardless. After months of research and experimentation using different software, I have successfully designed a three-way speaker to work for my needs.

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Driver Choice

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Box Design

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Crossover

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